

Movie Knowledge Graph Analysis Using Neo4j and the MovieLens Dataset

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Abstract. In this study, a knowledge graph is built and analyzed on MovieLens 100K using Neo4j. There are three node types used in this graph, namely User, Movie, and Genre, which are related by RATED, HAS_GENRE, and TAGGED relationships. Five research questions were formulated and answered via Cypher Queries, out of which, three research questions were answered via graph visualization using Neo4j while the other two are presented in tabular form validated using Power BI.

Keywords: Knowledge Graph · Neo4j · MovieLens · Cypher · Graph Analysis · Data Visualization · Power BI

1 Introduction

Various digital media providers such as Netflix, Amazon Prime Video, and MovieLens create huge volumes of data that capture user activity, ratings, reviews, tags, and user watch history. It is crucial to extract relevant information from this user-generated data to create recommendation engines and understand user behavior.

A traditional approach based on relational databases and SQL queries can easily aggregate user-generated data into averages and statistics, but cannot be used for solving complicated multi-hop relationships. For example, determining the existence of implied relations among users, who share movie interests and have similar genre preferences, involves complex self-joins that are not only costly but also hard to implement.

Knowledge graph is an alternative model that represents entities as nodes and interactions as edges, allowing for the representation and traversals of complex relations. In this project, we build a knowledge graph using the MovieLens 100K dataset [1] and the Neo4j graph database [2]. The knowledge graph contains three node labels—User, Movie, and Genre—and relationships—RATED, HAS_GENRE, and TAGGED—between them. We use Cypher queries [3] to analyze the knowledge graph and answer five research questions, three generating graph visualization and the other two generating tabular outputs, cross-verified by Power BI.

The objectives of this project are:

1. **Identify Patterns:** Discover user behaviour and rating trends using both graph-based and tabular analysis.
2. **Map Relationships:** Construct a knowledge graph that reveals structural connections between users, movies, and genres beyond the scope of flat tables.
3. **Cross-Validate:** Demonstrate the complementary strengths of Neo4j and Power BI by independently verifying results across both platforms.

This report continues to be organized in the following manner. Section 2 discusses some related studies. Section 3 mentions the questions for research. Section 4 introduces data preprocessing. Section 5 deals with the design of the graph model. Section 6.1 explains data analysis and visualization

2 Related Work

Three different themes that are pertinent to the current study are addressed in this chapter: namely, knowledge graphs and graph databases, MovieLens Database, and recommendation systems and business intelligence systems for cross-validation.

2.1 Knowledge Graphs and Graph Databases

Knowledge graphs have gained popularity as the preferred model used to represent structured data as a network of entities. They have been employed in various industries since their inception by Google’s Knowledge Graph in 2012, such as medicine, finance, and fraud detection [7].

Neo4j is the most popular graph database system, providing native graph support, ACID transaction support, and the Cypher querying language [2, ?]. The declarative pattern-matching approach of Cypher queries allows multi-hop traversals to be expressed intuitively compared to SQL statements. Moreover, graph visualization capabilities can be achieved natively using the Neo4j browser.

Graph databases have been deployed in social network analysis, supply chain management, and entertainment fields such as actors’ collaboration network analysis. Nevertheless, research employing graph databases for the combination of constructing knowledge graphs based on open data sources along with conducting a scientific investigation to answer research questions has yet to emerge.

2.2 MovieLens Database and Recommendation Systems

The MovieLens dataset is the de-facto standard for evaluating recommendation algorithms at the GroupLens Research Lab [1]. The MovieLens dataset has been extensively used to evaluate collaborative filtering, content-based filtering, matrix factorization, and deep learning algorithms for recommending movies.

Collaborative filtering is the most widely used technique, wherein movies recommended are based on finding similar users and recommending movies that are liked by similar users. In conventional techniques, such as the SVD algorithm, the problem is reduced to operations involving matrix factorization; this process uses flat data representations without incorporating the structure among users, movies, and genres.

Graph-based recommendation systems propose using path traversal on a knowledge graph to make recommendations using paths such as $\text{User} \rightarrow \text{Movie} \leftarrow \text{User} \rightarrow \text{Movie}$ [8]. Such systems are inherently interpretable, use heterogeneous data, and scale better than conventional systems. However, most of the experiments done on the MovieLens dataset have used either relational databases or Python libraries for implementing recommendation models.

2.3 Business Intelligence Tools for Cross-Validation

Power BI is a popular business intelligence solution that provides interactive dashboards, calculations using DAX language, and drag-and-drop visualization capabilities [6]. Unfortunately, Power BI doesn't have any native features for handling graph structures, multi-hop queries, and visualization of nodes and edges. That is why Power BI should be considered an auxiliary tool to Neo4j and not its alternative.

3 Project Objectives and Research Questions

This project belongs to Option 1, Data Analysis Project, which involves working with knowledge graphs and Neo4j and other tools for data visualization. This project will follow the mandatory phases of data analysis projects—Data Collection and Preparation, Data Analysis and Visualization, and Insight Discovery—the construction of the knowledge graph and its analysis being non-trivial parts of the work.

3.1 Project Objectives

The main purpose of this project is the construction of a knowledge graph using the MovieLens 100K database and performing research of the relations between users, movies, and their genre based on this knowledge graph. In particular, the following goals will be pursued in this project:

1. **Find Patterns:** Find patterns in rating behavior by users, such as dependencies of rating behavior on the amount of activity of each user, whether there are any consistently more satisfying genres, and whether users exhibit some temporal bias in rating older and newer movies.
2. **Find Relations:** Create a knowledge graph that represents the connections between users, movies, and their genres as typed relationships and use it for finding patterns of structures among users, genres, and movies that cannot be revealed in standard flat table format.

3. **Cross-Validation and Comparison of Tools:** Show the unique strengths of graph analysis tools such as Neo4j and table-based business intelligence tools like Power BI by independently generating results using each tool and discussing the strengths and weaknesses of each platform in achieving its purposes.

3.2 Research Questions

In order to address the above goals, we propose five research questions. The research questions are strategically separated into two distinct groups: graph visualization questions, where the result will involve a graphical illustration that takes advantage of the structural nature of Neo4j, and tabular questions, where numbers will be used to demonstrate their results and then verified on Neo4j and Power BI.

Graph Visualization Research Questions These research questions are structured in a way that they will produce interactive graphs using Neo4j Browser.

- **RQ1: How do the most active users connect through shared highly-rated movies and genres?**
- **RQ2: What does a highly active user’s movie taste look like across genres?**
- **RQ3: How do the top 10 most-rated movies share genres with each other?**

Table Analysis Questions These are questions that result in structured quantitative answers that lend themselves to tabular and graphical representation. They are replicable using Cypher and DAX, meaning the answers should be verifiable in both platforms independently of each other.

- **RQ4: Which genres dominate user preferences, and how does genre popularity compare to genre satisfaction?**
- **RQ5: Are older movies rated higher than newer ones, suggesting survivorship or nostalgia bias?**

4 Data Collection and Preparation

4.1 Data Source

The MovieLens 100K dataset [1] provided by the GroupLens Research Lab is used in this study, consisting of around 100,000 ratings for more than 9,000 movies by about 600 users. There were three source files used in this experiment: the movies file, the ratings file, and the tags file. The only movies that have ratings are selected in order for all the nodes to have visible interaction data.

4.2 Data Preparation and Cleaning

Preprocessing of the initial data set involved several tasks: unifying identifiers and data types used in all data files, eliminating low quality or incomplete observations, converting flat file structure to node tables and relation tables for graph analysis purposes, and generating additional variables, such as years when movies were released.

Identifiers were converted to unique integer values, while rating values were converted to float type. Time stamps were converted to easily readable datetime objects. Duplicate movies were identified on the basis of their identifiers, while duplicate rating/Tag pairs were identified based on their compound keys.

4.3 Structural Transformation for Graph Modeling

The genre column in the movie table, which had been delimited by pipes, was expanded, allowing each genre to become an individual node related to movies through a HAS_GENRE relation. Movie release years were identified within titles using regular expressions and saved as a separate attribute called `year` on the Movie node.

The result of this process was six CSV files, comprising three tables for nodes (`users_node`, `movies_node`, `genres_node`), and three tables for relations (`ratings_rel`, `movie_genre_rel`, `tags_rel`).

4.4 Manual Repair and Data Quality Refinement

Other films had titles that didn't follow year conventions or were irregularly named. Films having just one rating and without tags were discarded as noise, while other films that showed some interaction were fixed manually. Genres such as IMAX were deleted, and "no genres listed" movies underwent further labeling, wherever feasible. Manual filtering was essential in this case because entities needed to be precisely defined in a graph setting.

5 Graph Model Design

5.1 Knowledge Graph Schema

The knowledge graph consists of three node types and three relationship types:

Node Types:

- **User** – 610 users identified by `userId`.
- **Movie** – 9,733 movies with properties `movieId`, `title`, and `year`.
- **Genre** – 19 genre categories identified by `name`.

Relationship Types:

- **RATED** – User to Movie, with `rating` (0.5–5.0) and `timestamp` properties. 100,827 edges.

- **HAS_GENRE** – Movie to Genre, indicating genre membership. 21,949 edges.
- **TAGGED** – User to Movie, with `tag` and `timestamp` properties. 3,683 edges.

The complete graph contains 10,362 nodes and 126,459 relationships, summarized in Table 1.

Table 1. Knowledge graph composition.

Component	Type	Count
User	Node	610
Movie	Node	9,733
Genre	Node	19
Total Nodes		10,362
RATED	Relationship	100,827
HAS_GENRE	Relationship	21,949
TAGGED	Relationship	3,683
Total Relationships		126,459

Uniqueness constraints were created on `userId`, `movieId`, and `genre name` to prevent duplicates and automatically generate indexes for query performance.

5.2 Rationale for Graph-Based Modeling

The graph database model was selected because of four reasons. First, the MovieLens data is relational in nature; users rate movies; movies belong to genres; and graph databases represent this as explicit edges between first-class entities instead of foreign key relationships. Second, research questions such as RQ1 involve traversing several edges in one direction, e.g., `User` $\rightarrow$ `Movie` $\leftarrow$ `User` $\rightarrow$ `Genre`, that are straightforward in Cypher but complicated to perform in SQL via multi-self join operations. Third, visual analytics capabilities of Neo4j database are required to solve RQ1, RQ2, and RQ3 by applying visual pattern finding techniques. Fourth, flexibility of graph database models will allow to extend the graph in the future with additional nodes such as Actors or Directors.

5.3 Analytical Role of the Graph Schema

Each component within the schema plays a unique role in the analysis process. For instance, the schema that consists of two users rating the same film is the foundation for analyzing similarities between users, which constitutes research question 1. Movie to Genre connections provide aggregation at the genre level (research question 4) while also showing clusters of highly-rated films (research question 3). The pathway from User through Movie to Genre allows one to analyze user preferences, which is research question 2.

6 Insight Discovery

This section presents the findings from each research question and describes the cross-validation process used to verify tabular results between Neo4j and Power BI.

6.1 Research Questions

RQ1: How do the most active users connect through shared highly-rated movies and genres? Through the graph visualization of RQ1, it became clear that the most active users on the platform do not operate in isolation as prolific raters, but are rather interlinked by a large network of popular movies rated by them. This is because users were selected on the basis of having rated certain movies with ratings equal to 4.0 or higher, with the resulting graph showing such users as nodes that were connected via their common Movie nodes.

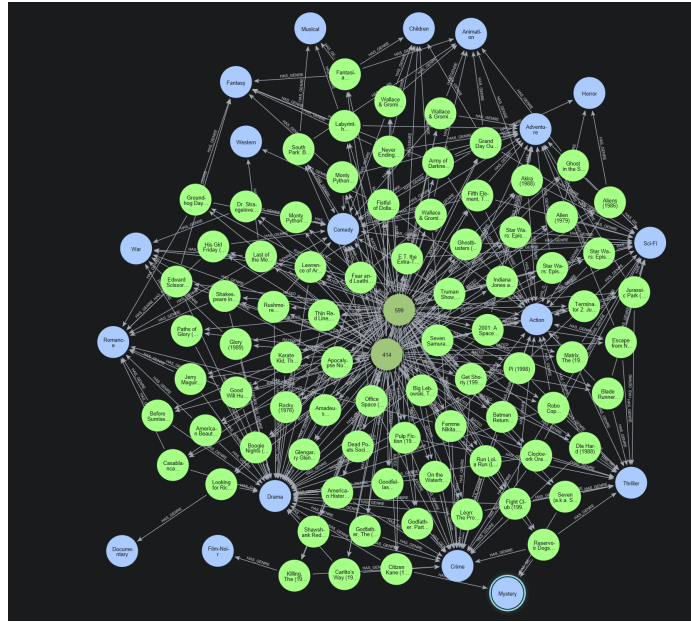


Fig. 1. RQ1: Graph visualization showing the most active users (orange) connected through shared highly-rated movies (blue), with genre hubs (green) as structural bridges.

Some notable insights from the visualization included: the first was that the movies shared between active user pairs were no mere coincidence but were critically praised and popular films, implying that users actively participate in forming a common collection of reputable media. Another insight was the role of

the Genre nodes, including Drama, Thriller, and Sci-Fi, which acted as structural intermediaries between user pairs, highlighting the fact that users are similar through their general affinity towards genres and not niche interests. Finally, there were differences between how connected each user pair was since some pairs overlapped in terms of sharing dozens of movies that received high ratings while other pairs exhibited less similarity.

The implication for recommendation engines of such an observation is the possibility of forming recommendation paths for active users through traversing the network with constraints for the path to contain only high rating movies.

RQ2: What does a highly active user’s movie taste look like across genres? The preference map drawn for the most active user had a unique star-like pattern in which the User was placed in the center with Movie nodes having ratings of 4.5 or higher radiating from the center with Genre nodes forming the ring around them.

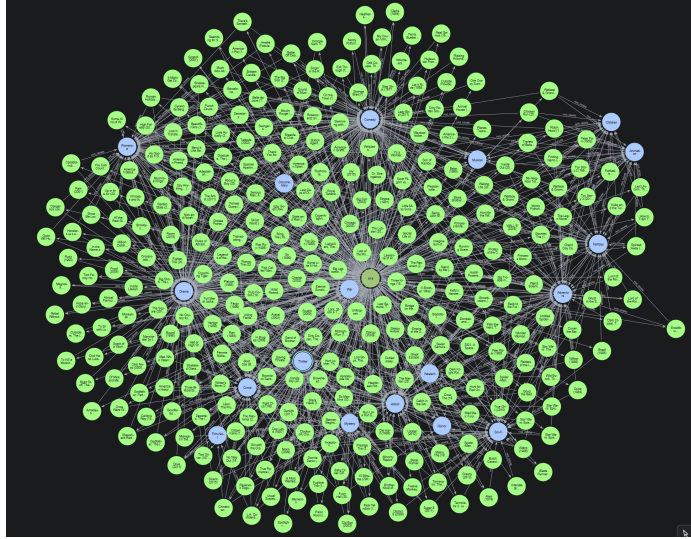


Fig. 2. RQ2: Preference map of the most active user (orange, center) radiating outward to highly-rated movies (blue) and their genre connections (green), revealing broad exploratory taste.

It was found out that the most active user is a seeker of diverse movies instead of a movie fan who prefers a particular genre. Films that received 4.5 stars or more belong to different genres, connecting to Drama, Comedy, Thriller, Action, Adventure, Sci-Fi, and other types of movies. But not all genres received equal attention from the user; some types, like Drama and Thriller, were more favored by the user compared to other kinds.

Moreover, it could also be seen in the graph that most of the movies rated higher by the user are multifaceted films because most of them are related to different genres at once, forming a spiderweb-like network. Instead of a single-starred connection, multiple genres connect with each Movie node, which might be the reason why the user has a wide range of interests—these films are mixed with several genres.

RQ3: How do the top 10 most-rated movies share genres with each other? One of the most informative outputs from this visualization exercise was that obtained when clustering the genres among the top 10 highest-rated films. The output showed 10 Movie nodes being connected with a few Genre nodes. From the graph, it was very clear to see which genres were central, while others were not so.

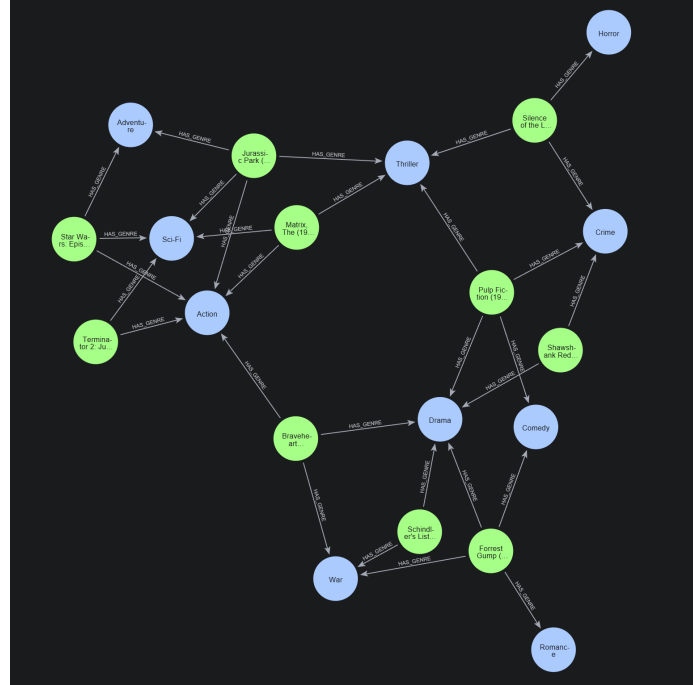


Fig. 3. RQ3: Top 10 most-rated movies (blue) clustered around shared genre nodes (green), with Drama and Comedy emerging as dominant hubs.

The main result obtained from the network graph analysis is the existence of certain central genres like Drama and Comedy which served as a hub for the largest proportion of top 10 movies. The presence of these genres in the middle part of the network graph could be observed because of their high connections

with other Movies nodes. Therefore, the movies with high popularity were usually found within broad genres which appeal to wide target audiences.

On the contrary, certain narrow genres, for example, Sci-Fi, Thriller, and Crime, could serve as secondary nodes in the network graph because they did not relate to all the top 10 movies. That is why these genres acted as differentiators rather than aggregators.

One of the fascinating aspects of movie structure in terms of genre is that some of the top 10 movies belonged to more than one genre at a time. If, for instance, a movie belongs to Drama and Thriller, then the film will have two connections with other films; thus, there will be multiple paths in the graph. It can be said that movie genres overlapping is one of the reasons why movies become popular because they will attract fans from both genres.

RQ4: Which genres dominate user preferences, and how does genre popularity compare to genre satisfaction? The genre analysis query returned results for all 19 genres in the dataset, revealing a clear distinction between genre popularity and genre satisfaction.

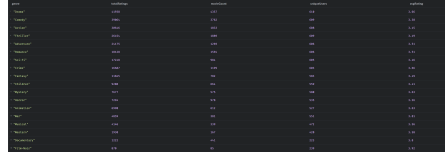


Fig. 4. RQ4: Genre popularity and satisfaction results.

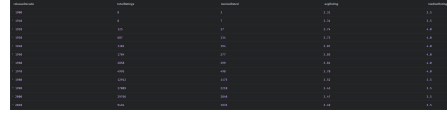


Fig. 5. RQ5: Ratings by release decade.

Drama, however, was the most popular genre according to the number of ratings, with Comedy, Action, Thriller, and Adventure following close behind. It is clear that these top five genres account for the overwhelming majority of all ratings in the platform. On the opposite end of the scale, niche genres like Film-Noir, IMAX, and Musical earned considerably lower numbers of ratings due to their relatively low number of viewers.

The mean rating per genre was a completely different story, however, and showed how some of the more popular genres were not necessarily the most satisfying ones. The mean ratings per genre revealed that several niche genres had mean ratings higher than the most popular genres, meaning that despite the small number of people who watched and rated these genres, they had better satisfaction levels with the movies they chose. This result is consistent with the hypothesis of self-selection, wherein users of these genres are usually more targeted, making their ratings more satisfactory.

The difference between popularity and satisfaction is relevant for recommendation engines since an algorithm based on recommendations of popular genres

alone would ensure that the largest number of interactions took place but would fail to exploit any niche genres that could have achieved more satisfaction.

RQ5: Are older movies rated higher than newer ones, suggesting survivorship or nostalgia bias? Temporal analysis involved sorting all the movies that had been rated based on their decades of release, and then calculating the mean and median rating for each decade. It was found that there was a continuous decline in mean rating from one decade to another.

Movies made in earlier decades (1920s to 1970s) had a considerably higher mean rating relative to movies made during the 2000s and 2010s. This trend held true for both mean and median rating measures, thus eliminating the risk that outliers might have caused the higher mean ratings for earlier decades.

There are two mutually reinforcing reasons why this happened. First, there is a selection effect due to survivorship bias. Movies made during the 1940s or 1960s that survived until today and still get rated on contemporary platforms are almost certainly the best movies produced during that decade. Poor-quality movies made during that period have not survived and thus have not made it into the data set, thereby driving up the mean rating for earlier decades.

Explanation two is nostalgia bias, whereby users tend to rate old movies more highly due to their cultural significance or historical value or because of nostalgia. For instance, movies such as *Casablanca* or *Godfather* have cultural significance that might cause unconscious exaggeration of the rating beyond what it would be rated were it made today.

The practical application of this result is that recommender systems need to take into account time bias while comparing the ratings of films over the years. For example, a movie produced in the 1960s that has a rating of 4.2 is not comparable to a modern movie that has an average rating of 3.8.

6.2 Cross-Validation (Neo4j vs Power BI)

To ensure the reliability and accuracy of the analytical findings, all five research questions were independently validated using Power BI alongside Neo4j. Both tools operated on the same underlying CSV data files, allowing direct comparison of results. For RQ2, RQ3, RQ4, and RQ5, the key metrics — including genre distributions, movie rankings, rating counts, and decade-level averages — matched consistently across both platforms, confirming the integrity of the data pipeline and the accuracy of the Cypher queries. For RQ1, the graph-based relationship visualization was performed exclusively in Neo4j, as this type of multi-hop user-movie-genre connectivity analysis is a graph-native capability not directly replicable in Power BI’s tabular model. However, the underlying data components — including the identity of the top 3 most active users — were independently verified in Power BI. This cross-validation process confirmed that the two tools produce consistent results where their capabilities overlap, while also highlighting the unique analytical strengths that graph databases bring to relationship-centric questions.

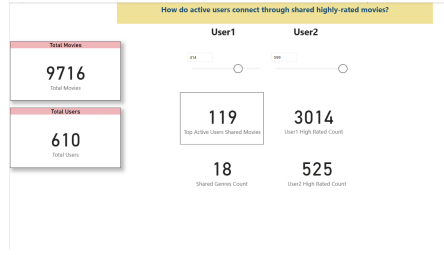


Fig. 6. RQ1: Shared movies between active users

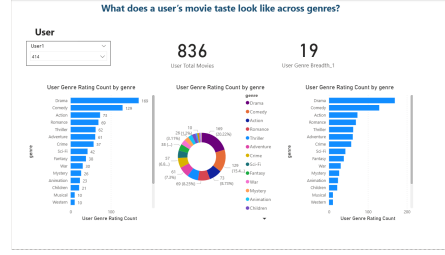


Fig. 7. RQ2: Users genre taste profile

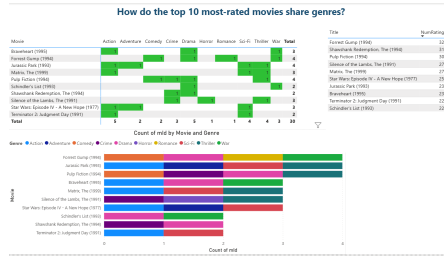


Fig. 8. RQ3: Top 10 movies genre overlap

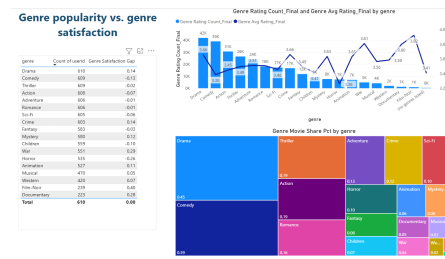


Fig. 9. RQ4: Genre popularity vs satisfaction dashboard

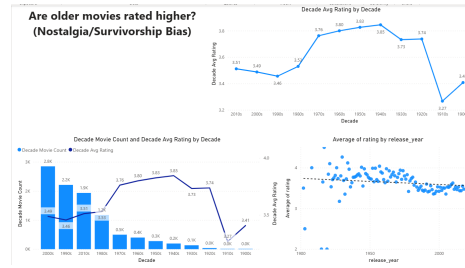


Fig. 10. RQ5: Decade-level nostalgia bias analysis

RQ1 Cross-Validation For RQ1, Neo4j reported 81 shared highly-rated movies between Users 414 and 599, while Power BI reported 119. This difference occurs because the Power BI ratings table contains genre-merged rows, creating multiple entries per rating for multi-genre movies. Neo4j stores each rating as a single relationship, producing accurate counts natively. Despite this numerical difference, both tools identified the same top 3 most active users (414, 474, 599),

confirming data pipeline consistency. This discrepancy highlights a key project finding: graph databases handle relationship-centric queries more naturally than tabular BI tools.

RQ2 Cross-Validation Power BI shows User 414 rated movies across 19 genres, with Drama (169) and Comedy (129) leading their preferences. Neo4j’s graph traversal for the same user confirms the same genre distribution and ranking order, with Drama and Comedy as the top two genres. Both tools identify User 414 as a broad explorer across nearly all available genres.

RQ3 Cross-Validation Both tools returned the same top 10 most-rated movies in the same order — Forrest Gump (329), Shawshank Redemption (317), Pulp Fiction (307), and so on. The genre overlap is also consistent: Drama appears in 5 of the 10 films, confirmed by both the Power BI matrix heatmap and Neo4j’s *HAS_GENREconnections*.

RQ4 Cross-Validation Drama leads with approximately 42,000 total ratings in both tools, while Film-Noir scores the highest average rating at 3.92 despite having the fewest ratings. The genre ranking by total ratings and by average rating matches across Neo4j and Power BI, confirming that genre popularity and genre satisfaction are inversely related in this dataset.

RQ5 Cross-Validation Both tools confirm that older decades receive higher average ratings — 1940s at 3.85, 1950s at 3.85, compared to 1990s at 3.43 and 2000s at 3.49. The decade-level averages match between Neo4j and Power BI with minor rounding differences. The Power BI scatter plot trend line supports a slight negative correlation, consistent with a survivorship bias pattern.

7 Non-Trivial Challenges

7.1 Data Transformation

The pipe separated column for genres needed a Python script to split it and generate rows separately, resulting in 21,949 rows being written to `movie_genre_rel.csv`. Release years had to be scraped out from the movie names using regex. Special cases like non-existent years and additional parentheses in the title were treated differently. Orphaned records were identified using SQLite.

7.2 Query Optimization

Timings for RQ1 first went into timeout while fetching the movies that top users agreed on without any filtering criteria. This was achieved by including (`r1.rating >= 4`), applying (`LIMIT 80`), and ensuring there were no duplicates (`u1.userId < u2.userId`). In order to extract the movie year for RQ5 from the string of titles, it took too long due to the size of 100,827 edges, thus we had to load it upfront as a property of nodes.

7.3 Cross-Platform Consistency

Type inference performed by Power BI necessitated manual fixing; for example, the `userId` was assumed to be numeric, and a decimal data type had to be defined explicitly for `rating`. The many-to-many relation between movies and genres, natively expressed using graph edges in Neo4j, had to be manually joined in Power BI as `ratings_rel` and `movie_genre_rel`.

8 Conclusion

In this case, the use of knowledge graphs was shown to be able to handle large-scale interaction data in the case of movie recommendations. Knowledge graph comprised 10,362 nodes and 126,459 relationships based on MovieLens 100K and was loaded into Neo4j Aura. This allowed five research questions regarding graph visualization and tabulation to be handled.

It is notable that graph visualization questions (RQ1–RQ3) generated some of the most interesting results throughout this project. To start with, RQ1 revealed that users create implicit communities through enjoyment of common high-rating movies, and genre nodes become structural links among the users. Second, RQ2 revealed that the user most active is diverse in terms of preferences for different types of movies and appreciates genre fusions. Third, RQ3 suggested that high-rating movies are centered around genre hubs such as Drama and Comedy; thus, genre diversity contributes to their popularity.

In turn, tabulation questions (RQ4–RQ5) provided some numerical results of analysis. RQ4 proved that the discrepancy exists between genre popularity and genre satisfaction, as less popular genres are rated higher compared to more popular genres due to selection biases. RQ5 found that there was always an age-related rating bias, as old movies tend to score higher.

The cross-validation results between Neo4j and Power BI showed the validity of all tabular results and showcased how the two tools complement each other. While Power BI is great at creating dashboards by aggregating data, Neo4j gives you insights into data structure through relationship traversing which is not possible with tabular data. The fact that you cannot perform graph queries (RQ1–RQ3) in Power BI makes their analytical capabilities complementary.

8.1 Future Work

There are multiple ways in which this project can be further developed. First, movie descriptions can be represented as vector embeddings and kept in a vector database for performing semantic searches and using RAG techniques in generating natural language recommendations. Neo4j’s GDS library can be used to conduct community detections using Louvain and PageRank algorithms. Relationships’ timestamp property can be used for performing temporal analysis on the evolution of users’ preferences. Lastly, we can enrich our graph with new entities Actor, Director, and Production Company with the help of IMDb and TMDb.

In summary, the project demonstrates that knowledge graphs present an entirely new approach to data analysis that unveils underlying structures and relationship-based information that are otherwise obscured through conventional tabular analysis. The synergy between Neo4j and Power BI presents an effective analytical model that can be applied to any field where entities interact.

References

1. Harper, F.M., Konstan, J.A.: The MovieLens Datasets: History and Context. *ACM Transactions on Interactive Intelligent Systems* **5**(4), Article 19 (2015)
2. Neo4j, Inc.: Neo4j Graph Database. <https://neo4j.com/> (2024)
3. Francis, N., et al.: Cypher: An Evolving Query Language for Property Graphs. In: *Proceedings of the 2018 International Conference on Management of Data*, pp. 1433–1445. ACM (2018)
4. GroupLens Research: MovieLens Datasets. <https://grouplens.org/datasets/movielens/> (2024)
5. Neo4j, Inc.: Neo4j AuraDB – Fully Managed Cloud Graph Database. <https://neo4j.com/cloud/aura/> (2024)
6. Microsoft Corporation: Power BI – Interactive Data Visualization. <https://powerbi.microsoft.com/> (2024)
7. Hogan, A., et al.: Knowledge Graphs. *ACM Computing Surveys* **54**(4), Article 71 (2022)
8. Guo, Q., et al.: A Survey on Knowledge Graph-Based Recommender Systems. *IEEE Transactions on Knowledge and Data Engineering* **34**(8), 3549–3568 (2022)